HISTOLOGY OF SALIVARY GLANDS

Learning objectives

At the end of the lecture the student should be able to know:

- Salivary glands and their location
- Histology of parotid gland
- Histology of submandibular gland
- Histology of sublingual gland.

Salivary Glands – Overview

- Salivary glands **are exocrine glands, glands with ducts, that produce saliva.**
- There are three major paired salivary glands: the submandibular, the sublingual and the parotid glands. They differ from one another in the relative abundance of serous and mucous acini, and in the length of the various kinds of ducts.
- The minor salivary glands are located in submucosa of different parts of oral cavity.



Functions Of Salivary Glands

- These glands produce a steady supply of saliva that trickles into the mouth to moisturize mucous membranes.
- Stimuli related to eating, such as the smell or appearance of food, activate an increased flow of saliva to meet the needs of mastication (chewing).
- Enzymes in saliva begin to break down foods to prepare them for digestion.
- Its high bicarbonate content buffers the oral cavity. It provides a medium for food materials to stimulate taste buds.
- It controls the bacterial flora by secreting lysozyme.
- The salivary glands also secrete IgA and potassium, and resorb sodium.

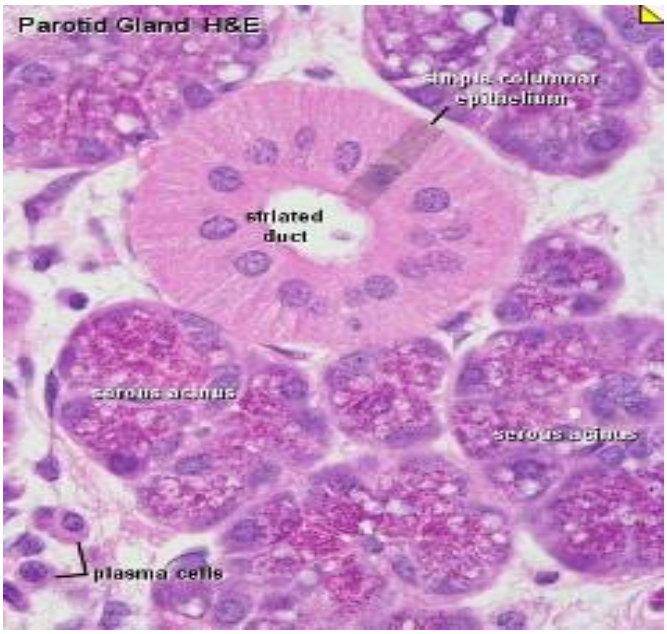
Histology of Salivary Gland

- These glands are divided into numerous lobules which contain the secretory units, which are called acini.
- Each secretory unit consists of acinar (secretory) cells and branching ducts.
- Ducts within lobules (*intralobular ducts*) connect with larger *interlobular ducts* which are found in the connective tissue between lobules.

Histology of salivary acini (secretory units)

- Unlike the parotid gland, the submandibular and sublingual glands possess both mucous and serous secretory cells.
- Mucous cells have a pale-staining, bubbly cytoplasm and nuclei that appear to be pushed against the basal cell membrane.
- In contrast, the *cytoplasm of serous cells contain distinctly eosinophilic zymogen granules and their nuclei are spherical in appearance.*

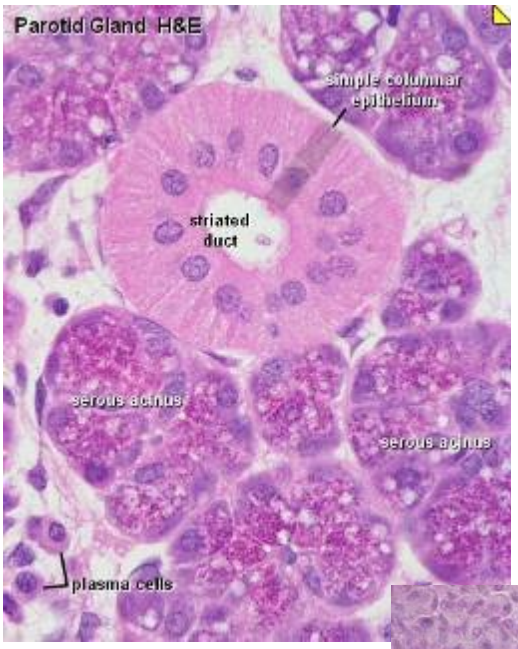
- The secretory cells are organized into secretory units or acini, and they are classified according to the cell types they contain:
- **serous acini** -contain only serous cells and are generally spherical
- **mucous acini** -contain only mucous cells and are generally tubular
- **mixed (or sero-mucous) acini** -contain a mixture of serous and mucous cells. In routinely-prepared tissue sections, the mucous cells swell and "push" the serous cells into a peripheral cap known as a serous demilune.

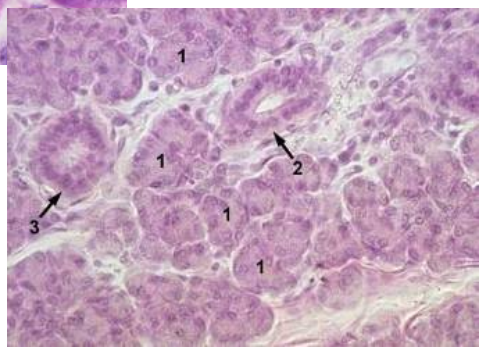


Parotid gland

- Largest.
- wrapped around the mandibular ramus, anterior and inferior to external ear
- Occupies the **parotid fascial space**
- Secretes saliva through **Stensen's duct** into the oral cavity, to facilitate mastication and swallowing.

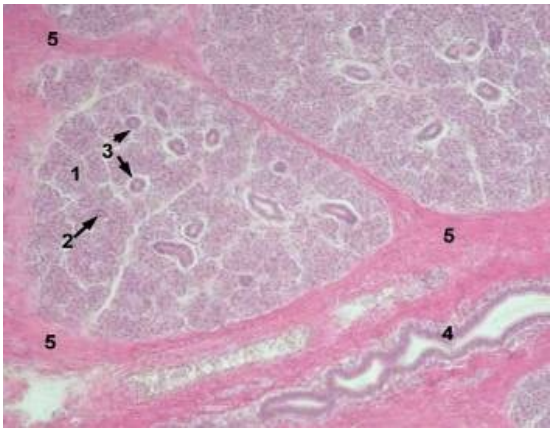
- There is abundance of fat in this gland: the amount of fat increases with age. Note that the parenchymal (secretory) tissue is divided into many lobules by the stromal (supportive) connective tissue.
- In this connective tissue stroma, notice the presence of blood vessels, nerves and large excretory ducts (interlobular).
- These interlobular ducts are lined by a pseudostratified columnar epithelium.
- In a well defined lobule the secretory acinar cells and ducts are seen .
- these acinar cells are fairly uniform in size, structure and staining, because they all produce proteinaceous (serous) secretion.
- With each lobule *are intralobular ducts which are lined by a* simple columnar epithelium of varying heights (becoming progressively taller toward the point at which the intraclobular ducts empty into interlobular ducts).
- The intralobular ducts are, in turn, further subdivided into striated and intercalated portions.
- The slender, intercalated ducts connect the secretory acini and striated ducts





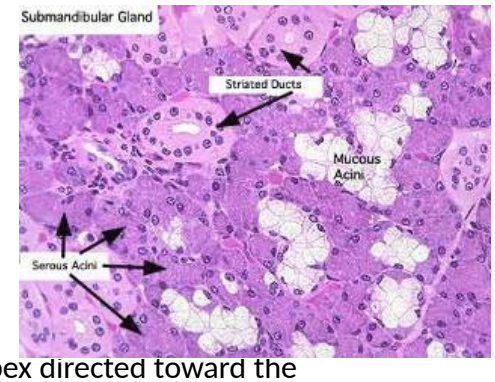
PAROTID SALIVARY GLAND
Stained with haematoxylin and eosin
1 - serous secretory units
2 - striated excretory duct
3 - interlobular excretory duct

PAROTID SALIVARY GLAND
Stained with haematoxylin and eosin
1 - serous secretory units (acini)
2 - intercalated excretory duct
3 - striated excretory duct
4 - interlobular excretory duct
5 - interlobular connective tissue septa



Submandibular gland

- Lying superior to the digastric muscles, each submandibular gland is divided into superficial and deep lobes.
- The superficial portion is larger.
- The deep portion is smaller.
- Secretions are delivered into the Wharton's ducts on the superficial portion after which they hook around the posterior edge of the mylohyoid muscle and proceed on the superior surface laterally.
- The ducts are then crossed by the lingual nerve, and ultimately drain into the **sublingual caruncles** on either side of the lingual frenulum along with the major sublingual duct (Bartholin).
- Submandibular gland is a **compound tubuloalveolar gland**.
- It is surrounded by a capsule of moderately dense connective tissue, from which septa divide the gland into lobes and lobules.
- The secretory end pieces consist of a more or less spherical mass of cells called an **acinus**).



- Cells of the acini appear triangular in sections, with their apex directed toward the lumen, and their base resting on a basement membrane.
- They secrete their product in a merocrine fashion into the lumen.

- Contractile cells called **myoepithelial cells** or **basket cells** lie between the basement membrane and the plasma membrane of the secretory cells.
- They are also found in the proximal part of the duct system.
- Myoepithelial cells posess many actin-containing microfilaments, which squeeze on the secretory cells and move their products toward the excretory ducts.
- Acini are
- **Serous**
- secretion of serous cells is thin, watery and proteinaceous. Serous cells have a rounded nucleus and secretory granules in their cytoplasm. They are joined near their apical surfaces by junctional complexes
- **Mucous**.
- Mucous cells secrete a *viscous*, *glycoprotein*-rich product, which is stored as mucinogen granules. The nuclei are typically flattened against the base of the cells (unless the cells have just discharged their contents, in which case they look more like serous cells).
- Mucous cells typically look pale and empty in standard histological sections, because their granules are lost during preparation.
- The submandibular gland of humans is predominantly serous. Its mucous acini are quite frequently capped with a **serous demilune**, a crescent of serous cells around one or more of their surfaces.
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3 types of ducts in the submandibular gland:

Intercalated

- Intercalatd ducts are slender ducts continuous with the terminal acini, and lined with flat, spindle-shaped cells

Secretory (**striated**)

- Secretory ducts have eosinophilic cuboidal to columnar cells with basal striations. These result from infoldings of the basal membranes in which are found many mitochondria.
- Both intercalated and secretory ducts are found within the parenchyma of the gland and are therefore **intralobular ducts**.

Excretory.

- The largest ducts are the excretory ducts. They are found in the connective tissue septa, and are therefore **interlobular ducts**. They ultimately connect with the oral cavity. Their epithelium is variable, it can be simple cuboidal, stratified cuboidal, stratified columnar or pseudostratified. Near the oral cavity, it becomes stratified squamous.

Submandibular gland

Stained with haematoxylin and eosin

1 - serous secretory unit
2 - mixed secretory unit
3 - intercalated excretory duct

4 - striated excretory duct
5 - interlobular excretory duct
6 - interlobular connective tissue septa
7 - mucous part of mixed secretory unit
8 - serous part (serous demilune) of mixed secretory unit

Sublingual gland

- They lie anterior to the submandibular gland under the tongue, beneath the mucous membrane of the floor of the mouth.
- They are drained by 8-20 excretory ducts called the **ducts of Rivinus.**
- The largest duct, the sublingual duct (of Bartholin) joins the submandibular duct to drain through the sublingual caruncle.
- The sublingual gland consists mostly of **Mucous** acini capped with serous demilunes and is therefore categorized as a mixed gland.
- Most of the remaining small sublingual ducts open separately into the mouth on an elevated crest of mucous membrane, the plica fimbriata, formed by the gland and located on either side of the frenulum linguae.
- *The **chorda tympani** nerve (from the facial nerve via the submandibular ganglion) is secretomotor to the sublingual glands.*

Sublingual gland
Stained with haematoxylin and eosin
1 - mucous part of mixed secretory unit
2 - serous part (serous demilune) of mixed secretory unit
3 - serous secretory unit
5 - myoepithelial cells
6 - intercalated excretory duct
7 - interlobular excretory duct
8 - interlobular connective tissue septa

Autonomic innervation

Parasympathetic

- Abundant,
- watery saliva
- Amylase down

Sympathetic

- Scant, viscous
- saliva
- Amylase up